



HORIZON POLICY

TEACHING ROBOTS TO WORK IN ANY LABORATORY

General Purpose Gantry Robotics Powered by Vision-Language-Action Models

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THE DAY WE FOUND THE PROBLEM

It Started Inside A Real Medical Laboratory

During visits to diagnostic laboratories, we observed a surprising reality.

Highly trained technicians were spending hours performing repetitive material handling tasks:

- Picking test tubes
- Scanning barcodes
- Moving samples between machines
- Organizing racks
- Sorting completed and pending tests

Labs wanted automation, but every solution required **custom engineering per workflow**.

The result: Expensive robots solving one task while humans stay the bottleneck.



Repetitive sample handling in diagnostic laboratories

THE REAL PROBLEM

Laboratories Don't Need Another Fixed Robot

CURRENT AUTOMATION SYSTEMS ARE:



Hardcoded



Workflow-specific



Expensive to deploy



Difficult to adapt

THE REALITY:

Today: Robot scans test tube → \$150K invested



Tomorrow: Need another robot for:

Refrigerator handling · Centrifuge loading · Sample transport · Machine feeding

Every workflow = a new engineering project

The industry is building **more legacy systems** instead of eliminating them.

OUR INSIGHT

The Problem Is Not Robotics

The Problem Is Adaptability

Humans succeed because they **understand goals**. A technician can walk into a new laboratory and immediately perform useful work.

Traditional robots cannot.

We believe laboratories need a robot that **understands instructions, observes the environment, and adapts its actions**.





OUR SOLUTION

A General Purpose Gantry Robot Powered By Vision-Language-Action Models

CORE COMPONENTS:



GANTRY PLATFORM

Industrial-grade precision positioning system



ROBOTIC ARM

Adaptive end effector for material handling



VISION SYSTEM

Multi-camera perception and scene understanding



VLA MODEL

Vision-Language-Action foundation model

NATURAL LANGUAGE INSTRUCTIONS:

“ Pick up the completed blood sample and place it in rack B.

“ Move all samples awaiting centrifugation to tray 3.

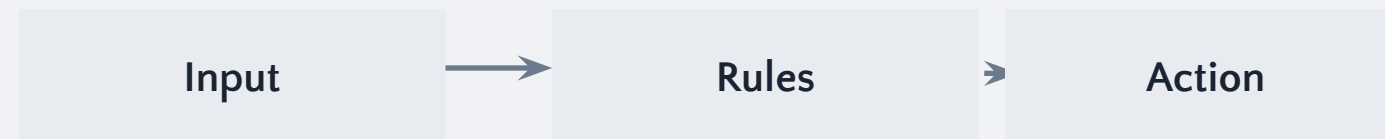
“ Retrieve the green-capped tube and place it in cold storage.

The robot learns policies rather than fixed procedures.

WHY VLA CHANGES EVERYTHING

From Programming Tasks To Teaching Intent

TRADITIONAL AUTOMATION

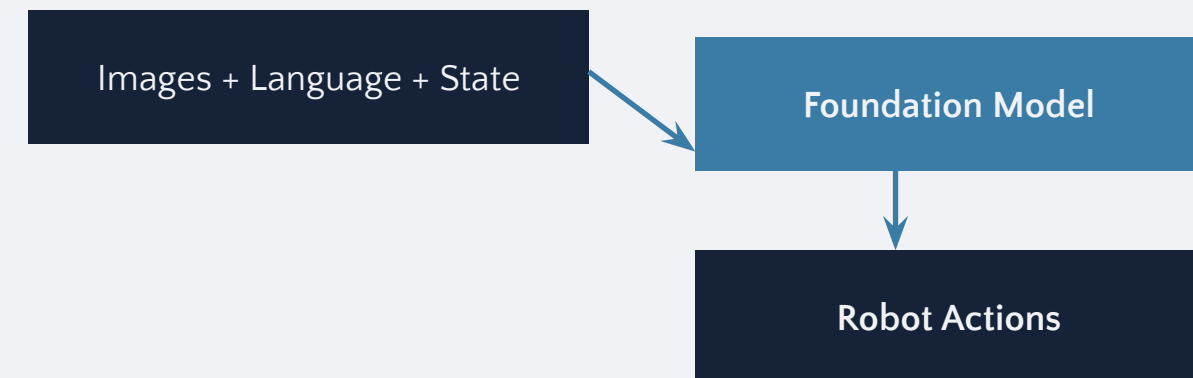


Requires engineers to manually define every behavior. Each new task needs custom programming.

KEY BENEFITS:

-  Adaptable workflows
-  Natural language instructions
-  Reduced engineering effort
-  Continuous improvement

VISION-LANGUAGE-ACTION



The model learns to map visual observations and natural language to physical actions.





TECHNICAL ARCHITECTURE

Horizon Policy Stack



PERCEPTION LAYER

Multi-camera system · Barcode recognition · Object detection · Scene understanding



FOUNDATION LAYER

Vision Language Action Model · Policy learning · Task planning · Semantic understanding



EXECUTION LAYER

Gantry controller · Robotic arm · Motion planning · Safety systems



FEEDBACK LAYER

Continuous learning · Human correction · Task logging · Dataset generation

INITIAL USE CASE

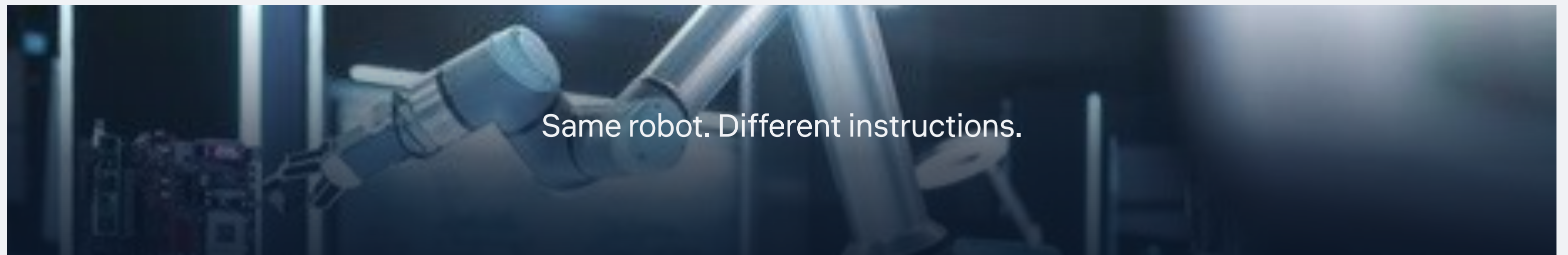
Medical Laboratory Automation

PHASE 1 CAPABILITIES

-  Test tube pickup
-  Barcode scanning
-  Rack placement
-  Sample sorting
-  Tray management

FUTURE TASKS

-  Refrigerator handling
-  Centrifuge loading
-  Machine feeding
-  Inventory movement
-  Multi-step workflows



WHY WE START WITH GANTRIES

The Fastest Path To Physical Intelligence

ADVANTAGES:



Controlled workspace



High precision



Lower safety risk



Easier deployment



Lower hardware cost

ROADMAP:



MARKET OPPORTUNITY

Laboratory Automation Is Growing Rapidly

MARKET DRIVERS:



TECHNICIAN SHORTAGES
Skilled lab staff increasingly scarce



RISING TEST VOLUMES
Diagnostic demand surging globally



RISING COSTS
Operational expenses increasing



24/7 PROCESSING
Demand for round-the-clock ops

LONG-TERM EXPANSION:

Medical laboratories are only the starting point. The same architecture expands into:





Pharmaceutical



Manufacturing



Warehouses



Research Labs



Industrial Auto

WHY US

TEAM BACKGROUND

-  Autonomous systems development
-  Computer vision
-  Robotics
-  Machine learning
-  Formula Student Driverless experience

CURRENT WORK

-  Vision-Language-Action systems
-  Gantry robotics
-  Autonomous perception
-  Real-world laboratory workflow analysis

We are building directly from field observations rather than hypothetical use cases.





THE VISION

Today: Robots are programmed.

Tomorrow: Robots learn.

Horizon Policy aims to create a general-purpose robotic workforce capable of understanding instructions and adapting to new environments.

Our first step is medical laboratories. Our long-term goal is physical intelligence that can operate anywhere humans perform repetitive tasks.

THANK YOU
